16. TESTING:

A. PRINTED WIRING BOARD AND TEST COUPON TO BE DESIGNED TO THE REQUIREMENTS OF IPC-2221 CLASS 3 AND TESTED PER IPC-6012B CLASS 3 TYPE 3.

B. ELECTRICAL TEST SHALL BE PERFORMED ON 100% OF THE DELIVERABLE PRODUCT. THE ELECTRICAL PROGRAM SHALL BE EXTRACTED FROM GERBER DATA OR IPC-356 FILE.

C. FOREIGN INCLUSIONS ALLOWED: 2 PER SIDE OF BOARD.

D. LAYER TO LAYER MIS-REGISTRATION ALLOWED: MAX .005. TOTAL OF .014 OVER ALL LAYERS.

E. NO OUTGASSING MASS LOSS TEST REQUIRED.

F. DIELECTRIC WITHSTANDING VOLTAGE (HIPOT) LAYER TO LAYER: 750VDC +15 -0 FOR 30 SECONDS. TO BE DONE ON TEST COUPON.

G. RATING: MAXIMUM RATED VOLTAGE: 5VDC.
TESTING TO BE AT DOUBLE THE MAXIMUM RATED VOLTAGE.

17. TEST COUPONS TO BE INTEGRATED INTO PANEL PER IPC-2221 CLASS 3 ARE TO BE ADDED BY MFG. TEST COUPONS TO BE TRACEABLE TO LOT AND PWB SERIAL NUMBER. TEST COUPON TO BE LOCATED WITHIN .5 INCHES OF ANY BOARD ON THE PANEL.

18. BOARD MANUFACTURER IS TO BE ON THE QPL FOR MIL-PRF-31032/1 OR MIL-PRF-55110.

9. HOLES:
DIAMETERS AND LOCATION TOLERANCES SHALL BE IAW IPC-2222, IPC-2221 CLASS 3 LEVEL B AND ASME Y14.5M-1994.
ALL HOLES ARE LOCATED

⌖	Ø.008 Ⓜ	A	B Ⓜ	C Ⓜ

AND ARE SPECIFIED IN THE ACCOMPANYING DRILL FILE.
FEATURES ARE DEFINED FROM DATUM B.

10. DRILL PER SUPPLIED FILE: S1101EFL3215_REV_B.PTH

11. ELECTRONIC DATA FILES:
GERBER FILES (IN EIA RS-274X) AND IMAGE FILES (PDF) ARE LISTED IN THE README.TXT FILE AND ARE SUPPLIED IN THE APPROPRIATE MEDIA (S1101EFL3215_REV_B_10-21-2014.ZIP)

12. BARE BOARD IDENTIFIED WITH PN AND ARTWORK REVISION. ARTWORK IMAGES ARE VIEWED THROUGH LAYER 1 (PRIMARY SIDE). CURRENT ARTWORK REVISION (CONFIG IDENT LETTER) IS LOCATED ON THE ARTWORK AND SECONDARY (BOTTOM) SIDE OF BOARD IN ETCH.

13. SOLDER MASK:
APPLY LPI (LIQUID PHOTO IMAGEABLE) PERMANENT GREEN SOLDER MASK ON PRIMARY (TOP) AND SECONDARY (BOTTOM) SIDE BOARD OVER BARE COPPER PER IPC-SM-840 CLASS H. FOR SOLDER MASK USE ARTWORK MASTER SUPPLIED, LISTED IN THE README.TXT FILE.

14. FINAL FINISH:
EXPOSED COPPER AREAS SHALL BE PRE-TINNED WITH A SINGLE PASS HOT AIR SOLDER LEVELING (HASL) PROCESS. THE PREFERRED HASL PROCESSING METHOD IS A HORIZONTAL PLANE HOT AIR SOLDER LEVEL UTILIZING A PROCESSING ANGLE OF 45 DEGREES.
PAD TO PAD THICKNESS UNIFORMITY SHALL BE MAINTAINED.
HASL MINIMUM THICKNESS SHALL BE .0003 INCHES.
HASL MAXIMUM THICKNESS SHALL BE .001 INCHES.
HASL CO-PLANARITY SHALL BE +/- 1000 MICRO INCHES.

15. MARKING:
SILKSCREEN, LEGENDS AND OTHER NOMENCLATURE ON TOP OF THE SOLDER MASK USING AA56032-IIWHT WHITE EPOXY INK PER MFG INSTRUCTIONS AND MIL-STD-130. FOR SILKSCREEN USE ARTWORK MASTER SUPPLIED, LISTED IN README.TXT FILE.
INDICATE THE FOLLOWING:
MANUFACTURER ID
DATE CODE
SERIAL NUMBER

NOTES: UNLESS OTHERWISE SPECIFIED

1. NO CHANGES BEYOND CUSTOMARY AND STANDARD BUILD PRACTICES SHALL BE MADE WITHOUT PRIOR APPROVAL FROM ZIN TECHNOLOGIES.

2. PRIMARY SIDE OF BOARD SHOWN.

3. FABRICATION SHALL BE IAW IPC-2221 AND IPC-2222.

4. INSPECTION AND ACCEPTANCE CRITERIA PER IPC-6012B CLASS 3 TYPE 3.

5. MATERIAL:

A. GLASS CLOTH EPOXY LAMINATE FLAME RESISTANT PER IPC-4101/24. COPPER ELECTRO DEPOSITED HIGH TEMPERATURE TYPE H, THICKNESS TOLERANCE CLASS B AND SURFACE QUALITY CLASS A. SEE LAMINATE STACK UP FOR NORMAL FOIL WEIGHT, UNLESS OTHERWISE INDICATED, MINIMUM DIELECTRIC SPACING BETWEEN CONDUCTIVE LAYERS SHALL BE .0035 MINIMUM AFTER LAMINATION.

B. BONDING SHEET: RESIN PRE-IMPREGNATED GLASS FABRIC PER IPC-4101/24.

C. COPPER FOIL: ELECTROPLATED COPPER SHEET PER IPC-MF-150, HTE-TYPE E. .5 OZ BEFORE PLATING MAY BE SUBSTITUTED FOR IPC-4101/24 MATERIAL FOR EXTERNAL LAYERS.

6. CONDUCTORS:
DESIGNED MINIMUM ACCEPTABLE CONDUCTOR WIDTHS AND SPACING AS FOLLOWS:

	INTERNAL	EXTERNAL
CONDUCTOR WIDTH ≤ TO .013	+.001 -.002	+.003 -.003
CONDUCTOR WIDTH > TO .013 AND < .025	+.002 -.004	+.004 -.004
CONDUCTOR WIDTH ≥ TO .025	+.004 -.006	+.008 -.006

WITH MINIMUM SPACING OF .010.
MINIMUM CONDUCTOR WIDTH OF .012.

7. MINIMUM INSULATION DISTANCE FROM ANY ACTIVE CIRCUITRY TO THE BOARD EDGE SHALL BE .019.

8. MINIMUM ANNULAR SOLDER RING SHALL BE PER IPC-6012, CLASS 3. THERE SHALL BE A MIN. OF .002 EXTERNAL AND .001 INTERNAL OF COPPER PAD ANNULAR RING, NOT INCLUDING HOLE PLATING MATERIAL, AROUND DRILL HOLES FOR ELECTRICAL CIRCUITS.

REV	DESCRIPTION	APPROVED
B	REVISED PER ACME-ECO-314 AND ACME-ECO-327	DAS 10/17/14 TAG 10/20/14 RJP 10/16/14
A	REVISED PER ACME-ECO-149	MH 12/17/13 TG 12/18/13 RP 12/20/13

REVISION HISTORY

.
S1101EFA3215
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
.X ± 0.03
.XX ± 0.01
.XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 10/21/14
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 12/24/12	QUALITY	D. CHOVAN 12/26/12
REVIEWER	M. HOMKO 12/27/12	MATERIALS	M. WILKINSON 1/2/13
MECHANICAL	A. DRAKE 1/3/13	PROJ LEAD	B. BOROWSKI 1/3/13
ELECTRICAL	T. GOBELI 1/3/13	-	-
STRUCTURAL	W. DIAL 1/3/13	-	-
SAFETY	W. BORRELLI 12/26/12	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
DELIVERY CODE: SPACE110

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
ACME
MOTION CONTROL BOARD INTERFACE
FABRICATION DRAWING

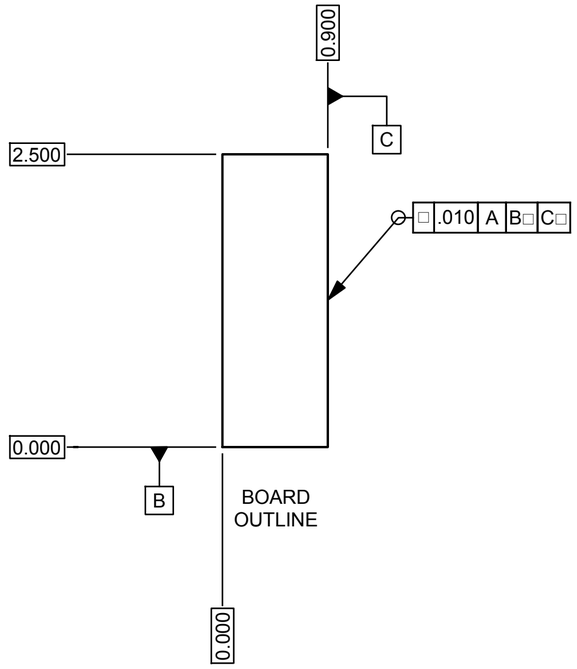
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C	0SG68	S1101EFF3215	B

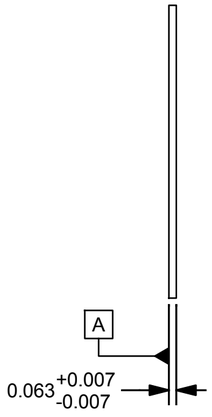
SCALE: NONE | INITIAL REL DATE 1/3/13 | SHEET 1 OF 2

RELEASE LEVEL: .
MODEL NAME: S1101EFA3215_REV_B.bpd

BLUEPRINT
VERSION 3.0

REV B
SH 1
DWG NO S1101EFF3215





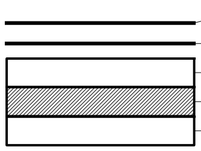


S1101EFL3215 REV B DRL

Through Holes				
Symbol	Quantity	Diameter	Tolerance	Plated
‹›	28	0.036	+0.004 / -0.003	Yes
A	8	0.039	+0.004 / -0.003	Yes

DRILL PATTERN

LAYER STACK-UP



SILKSCREEN (TOP)	FILE:	S1101EFL3215_REV_B.SST
SOLDERMASK (TOP)	FILE:	S1101EFL3215_REV_B.SMT
LAYER 1 (Signal) 0.0014 Cu (1.0 OZ)	FILE:	S1101EFL3215_REV_B.L 1
Substrate 0.062		
LAYER 2 (Signal) 0.0014 Cu (1.0 OZ)	FILE:	S1101EFL3215_REV_B.L 2
SOLDERMASK (BOTTOM)	FILE:	S1101EFL3215_REV_B.SMB
SILKSCREEN (BOTTOM)	FILE:	S1101EFL3215_REV_B.SSB

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFF3215	B
SCALE: NONE	INITIAL REL DATE 1/3/13	SHEET 2 OF 2	

DWG NO S1101EFF3215 REV B SH 2

BLUEPRINT
VERSION 3.0